

Unsupervised Anomaly Detection in Time Series with Evolving Spiking Neural Networks and Empirical Mode Decomposition

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Research Objective

The objective of this research was to develop and validate a novel ensemble-based time series anomaly detection method employing spiking neural networks in conjunction with CEEMDAN time series decomposition.

In order to achieve this, we:

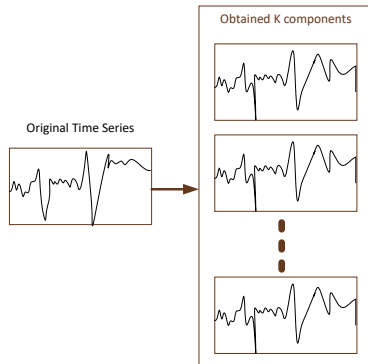
- 1 Conducted time series decomposition using the CEEMDAN method on data from the Numenta Anomaly Benchmark [Ahm+17a].
- 2 Designed two ensemble strategies for anomaly detection: one based on **majority voting** and the other on **weighted voting**.
- 3 Implemented a proposed ensemble consisting of Online Evolving Spiking Neural Networks (OeSNN), which has proven highly effective in detecting anomalies [Mac+21].

The CEEMDAN Time Series Decomposition

As a decomposition method, we have selected the Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEMDAN).

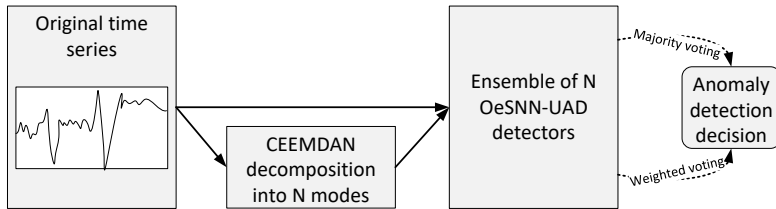
This method decomposes a single time series into a set of modes, each preserving two conditions:

- The number of extrema and the number of zero crossings must be equal or differ by at most one.
- The mean value of the upper and lower envelopes must be zero everywhere.



The Proposed Ensemble Model

- 1 The ensemble method is illustrated in the figure below.
- 2 Uses CEEMDAN to decompose time series.
- 3 Set of detectors for anomaly identification.
- 4 Each series is analyzed by a dedicated network.
- 5 Final anomaly detection via consensus (majority or weighted voting).
- 6 Enhances accuracy and robustness in detection.



Detailed Voting Strategies in Anomaly Detection

Ensemble Voting Strategies

Given a time series $X = [x_1, x_2, \dots, x_t, \dots, x_T]$ and a set of detections at time t , $D_t = [d_{0,t}, d_{1,t}, \dots, d_{N,t}]$ where $d_{i,t} \in \{0, 1\}$:

1 Majority Voting:

- An anomaly is reported if more than half of the detectors ($\frac{|D_t|}{2}$) identify an anomaly.
- **Mathematical Condition:** $|\{d \in D_t : d = 1\}| > \frac{|D_t|}{2}$.

2 Weighted Voting:

- Each detector's vote is weighted, typically based on the reliability or performance of the detector.
- Weights (w_i) may be assigned based on the squared reverse index of the detector's target CEEMDAN component.
- **Mathematical Condition:** $\sum_{i=0}^N w_i \cdot d_{i,t} > \frac{\sum_{i=0}^N w_i}{2}$.

Implementation available at

<https://pypi.org/project/OeSNN-AD/>.

Online Evolving Spiking Neural Networks

Overview

A streaming data anomaly detector using an evolving spiking neural network (OeSNN-UAD), focused on real-time classification and adaptive learning.

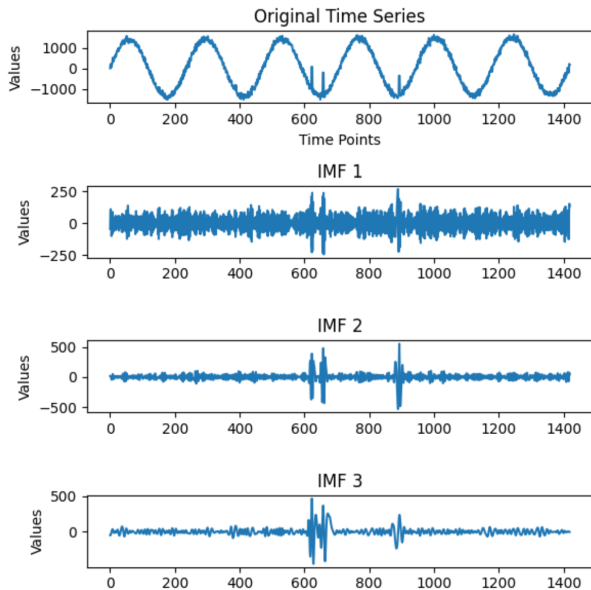
Architecture

Integrates modified OeSNN with modules for output generation, anomaly classification, and dynamic value correction, enhancing detection accuracy.

Operational Phases

- **Detection Phase:** Updates input window, classifies values based on neuron firing, and determines anomalies using prediction errors.
- **Learning Phase:** Adds and adjusts candidate neurons for refined future response based on incoming data.

CEEMDAN Time Series Decomposition Example



Original time series and its first three IMFs from CEEMDAN decomposition. Anomalies at points 600, 650, and 900 are distinctly visible in both the series and modes.

Experiments

Dataset Overview

Utilized the **Numenta Anomaly Benchmark (NAB)** dataset [Ahm+17b]:

- Over 50 real-world and artificial time series datasets.
- Features a scoring system that prioritizes timely and accurate anomaly detection.
- Designed for a variety of streaming data contexts.
- To evaluate anomaly detection we applied F1-measure.

Major Achievements

- **Weighted voting** by the OeSNN-UAD ensemble outperformed other methods in key categories like **Real Traffic** and **Artificial with Anomaly**.
- Demonstrated superior effectiveness over majority voting, particularly in environments sensitive to data distortion.

Conclusions of the Study

Innovative Approach

We introduced a new method for unsupervised anomaly detection combining **CEEMDAN decomposition** with an **ensemble of online evolving spiking neural networks**, demonstrating enhanced detection capabilities.

Empirical Results

Tested on the NAB dataset, our approach surpassed traditional detectors in certain categories, significantly benefiting from the **weighted voting strategy**.

Future Research Directions

- Explore additional datasets for broader validation.
- Compare CEEMDAN against FFT and other methods.
- Refine decision-making in the detection process.

- [Ahm+17a] Subutai Ahmad, Alexander Lavin, Scott Purdy, and Zuha Agha. “Unsupervised real-time anomaly detection for streaming data”. In: *Neurocomputing* 262 (2017). Online Real-Time Learning Strategies for Data Streams, pp. 134–147.
- [Ahm+17b] Subutai Ahmad, Alexander Lavina, Scott Purdya, and Zuha Agha. *Numenta Anomaly Benchmark*. Version 1.0. Nov. 1, 2017. URL: <https://github.com/numenta/NAB>.
- [Mac+21] Piotr S. Maciąg, Marzena Kryszkiewicz, Robert Bembienik, Jesus L. Lobo, and Javier Del Ser. “Unsupervised Anomaly Detection in Stream Data with Online Evolving Spiking Neural Networks”. In: *Neural Networks* 139 (2021), pp. 118–139. ISSN: 0893-6080. DOI: <https://doi.org/10.1016/j.neunet.2021.02.017>.